

National College of Ireland

Project Submission Sheet

Student Name:	CARIAPPA MALACHIRA KUSHALAPPA, ARUN ARUKANI DHAMOTHARAN, SUBHASH LORA JAT		
Student ID:	25144979, 24336572, 25171496		
Programme:	MSc. DATA ANALYTICS	Year:	2026-2027
Module:	BUSINESS INTELLIGENCE AND BUSINESS ANALYTICS		
Lecturer:	SEAN HEENEY		
Submission Due Date:	03-08-2026		
Project Title:	FC Highbury Business Intelligence & Analytics Solutions		
Word Count:	5914		

Academic Integrity Certificate Verification Code: fSoBtOWt4V, gjmEVdjOLy, 6lUoUeD6C8

I hereby certify that the information contained in this (my submission) is information pertaining to research I conducted for this project. All information other than my own contribution will be fully referenced and listed in the relevant bibliography section at the rear of the project.

ALL internet material must be referenced in the references section. Students are encouraged to use the Harvard Referencing Standard or appropriate referencing style supplied by the Library. To use other author's written or electronic work is illegal (plagiarism) and may result in disciplinary action. Students may be required to undergo a viva (oral examination) if there is suspicion about the validity of their submitted work.

Note: Academic Integrity Course Requirement – Starting Monday, 19th January 2026, all students are required to successfully complete the Epigeum Academic Integrity Course, which is available on Moodle. This is a self-paced online course consisting of 4 modules and will take approximately 2 hours to complete. Once you have successfully completed the course, you will receive a **Certificate & Verification Code**. This code must be included on the Project

Cover Sheet when submitting your work. For full details and guidance, please visit: <https://libguides.ncirl.ie/academicintegrity/epigeumforstudents>

<https://libguides.ncirl.ie/academicintegrity/epigeumforstudents>

Signature: CARIAPPA MALACHIRA KUSHALAPPA, ARUN ARUKANI DHAMOTHARAN, SUBHASH LORA JAT
.....
04-08-2026
Date:

PLEASE READ THE FOLLOWING INSTRUCTIONS:

1. Please attach a completed copy of this sheet to each project (including multiple copies).
2. Projects should be submitted to your Programme Coordinator.
3. **You must ensure that you retain a HARD COPY of ALL projects**, both for your own reference and in case a project is lost or mislaid. It is not sufficient to keep a copy on computer. Please do not bind projects or place in covers unless specifically requested.
4. You must ensure that all projects are submitted to your Programme Coordinator on or before the required submission date. **Late submissions will incur penalties.**
5. All projects must be submitted and passed in order to successfully complete the year. **Any project/assignment not submitted will be marked as a fail.**
6. **Please check that you read AI and Academic Integrity Acknowledgement Supplements in this document**

Office Use Only	
Signature:	
Date:	
Penalty Applied (if applicable):	

AI Acknowledgement Supplement

[Insert Module Name]

[Insert Title of your assignment]

Your Name/Student Number	Course	Date
25144979, 24336572, 25171496	BUSINESS INTELLIGENCE AND BUSINESS ANALYTICS	04-08-2026

This section is a supplement to the main assignment, to be used if AI was used in any capacity in the creation of your assignment; if you have queries about how to do this, please contact your lecturer. For an example of how to fill these sections out, please click [here](#).

AI Acknowledgment

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool
Claude	(1) Utilized to search for particular datasets	https://claude.ai/

Description of AI Usage

This section provides a more detailed description of how the AI tools were used in the assignment. It includes information about the prompts given to the AI tool, the responses received, and how these responses were utilized or modified in the assignment. **One table should be used for each tool used.**

[Insert Tool Name] Claude	
[Insert Description of use] Utilized to search for research papers	
[Insert Sample prompt] Find sports NFT related datasets	[Insert Sample response] CryptoSlam: This aggregator tracks historical sales volume, minting events, and rarity metrics specifically isolated for global sports NFT projects. [1] , [2]

Evidence of AI Usage

This section includes evidence of significant prompts and responses used or generated through the AI tool. It should provide a clear understanding of the extent to which the AI tool was used in the assignment. Evidence may be attached via screenshots or text.

Additional Evidence:

[Place evidence here]

Additional Evidence:

[Place evidence here]

FC Highbury Business Intelligence & Analytics Solutions



Presentation URL : <https://youtu.be/D3EeBpRjEQg>

Specification Report

(a) Background Information

FC Highbury is an iconic club established in 1886 in London, England. It is a historic club that has seen ups and downs and has been through most tiers of English football. It was once considered among the best clubs in the world but poor management & ownership led to a prolonged period of decline. Most recently, through a management restructure, clever investment and smart recruitment, the club has been on the up, gaining promotion back to the Championship and is financially stable. Key revenue streams for the club include broadcasting income, player sales, kits & accessories sales, tickets & membership sales. The club has a full squad of 28 players and 25 staff members that include coaching, ground & admin. The club also operates a highly rated youth academy setup with around 35 scholars.

Key figures at the club however, have identified that the club has hit a ceiling in terms of growth and see a growth potential much larger than what it currently is. This led them to hire us as consultants, to understand this club as a business, identify opportunities for growth, scale and set new business goals through analysis of existing information and implementing new solutions.

The Deloitte Football Money League identifies clubs as medium or large based on net revenue, and based on most recent information, it can be concluded that a medium tier club has a revenue of at least 200 million euros a year whereas Highbury FC has only achieved a net revenue of 210 million over the last 5 years. Medium to large size clubs earn at least 40% of their income through kit sales and sponsorships whereas that accounts for approx 22% of FC Highbury's revenue. In order to grow the business, a digital transformation in terms of a CRM enterprise system implementation is key as it can finally help us keep track of club supporters and track requests and issues and improve the overall customer experience which in turn grows the business. Highbury also faces competition from local rival clubs like NorthBridge FC, Red Oak United and Harbor Athletic for sponsorships and fanbases. Today, every single top club or a club that aspires to be at the top level has to understand the numbers around the club, what they mean and what can be optimized. As per research by PwC in 2023 on sports industry digitization, it shows that CRM implementation for fan engagement and BI analytics for financials are the two highest ROI investments that a sporting body can make.

(b) Business Goals

The following achievable business targets have been identified through analysis of visualizing FC Highbury's financial records and the the club's aspiration to operate as at least a medium level club -

	Business Goal	Current State	Target State	Timeframe	Owner
1	Increase matchday attendance	60% capacity	80% capacity	12 months	Commercial
2	Boost commercial revenue share	~22% of total income	40% of total income	18 months	Finance
3	Establish fan retention tracking via CRM	Untracked (no CRM)	≥70% retention rate	12 months	CRM
4	Reduce operational cost per match	~10%	7%	6 months	Operations
5	Boost matchday tickets & membership revenue	€105.68m last 5 seasons	+20%	24 months	Commercial

(c) Exploratory Data Analysis

EDA was done on the clubs finances, operations and player statistics over the last 5 seasons from 2021/22 to 2025/26. The key financial findings are described in this section.

Fig. 1 shows the income, expense, squad value over the past 5 seasons. The highest source of income has been through player sales which indicates that the club has a solid academy structure but also means that the club is unable to retain talented players unlike a medium sized club would be able to. Income through kits & accessories, tickets & memberships only accounts for 22% (169m) of the income which is a major point of focus as this should be at least at the 40% range for a club that is growing. Broadcasting income is at a healthy ~20% at 150m over the last five seasons. The club also makes a small portion of their income through renting out the stadium for other events (6.49% or 47m),

income received for the league position they finish in (10.58% or 76.6m) and income from sending players out on loan (3.9% or 28m).

The club has been progressing in the league and this is seen in the expense pie chart as player transfer fees accounts for 56.7% (291m over 5 seasons) of expenses and player wages and bonuses comes up to 27.74% (142m). This puts pressure on the club as it struggles to find finances for infrastructure, marketing. Misc expenses, IT services, loan repayment and staff wages accounts for the rest of it.

To help increase revenue and get into the NFT space for digital asset creation, we analyzed the Cryptoslam Sorare data that stores sales volumes, transactions and number of buyers information for the Sorare sports platform. Based on the trends over the last few seasons, it was found that NFT sales are on a downward trend and we decided not to pursue it further.

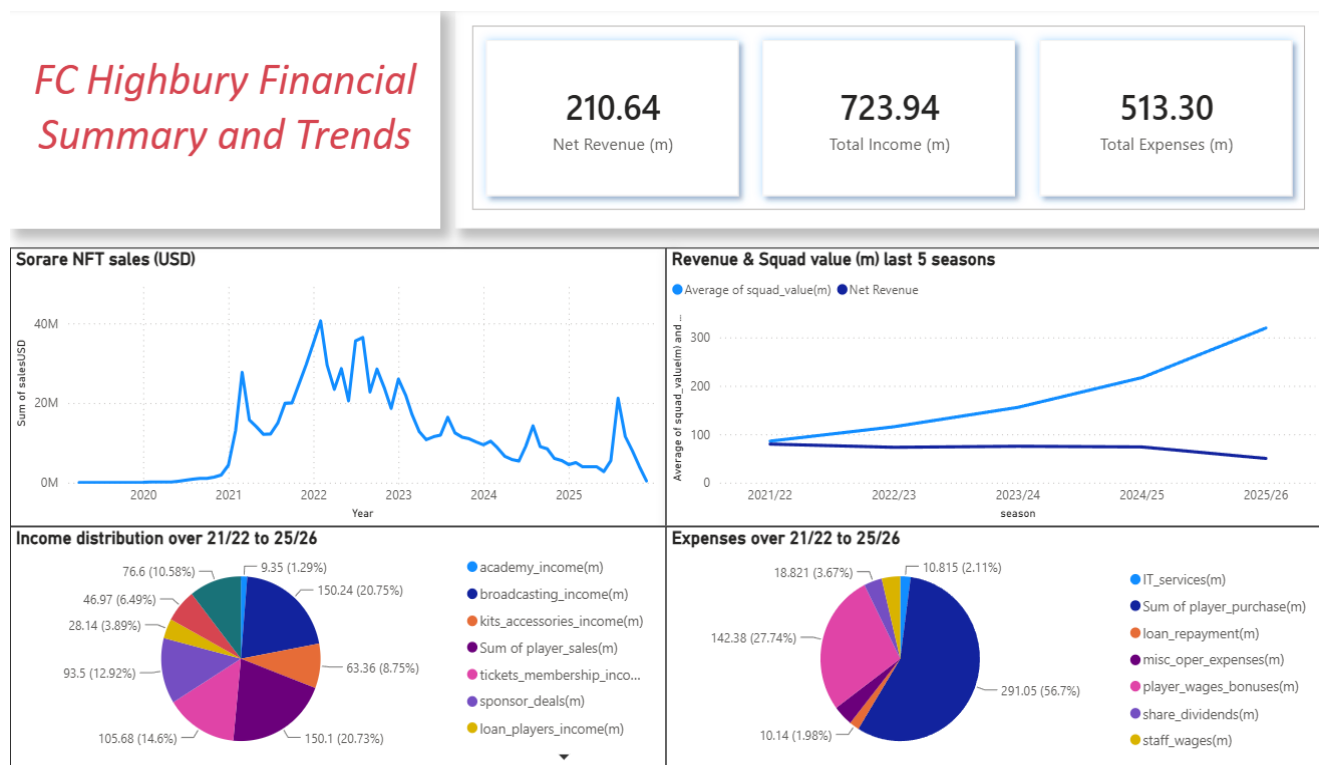


Fig. 1

Finally, from the line chart depicting the squad value and revenue in Fig 1, it can be seen that as the club progresses on the pitch with better a higher valued squad, the revenue has been stagnant and has tanked from 73.77m pounds in 24/25 to 49.99m pounds for the 25/26 season.

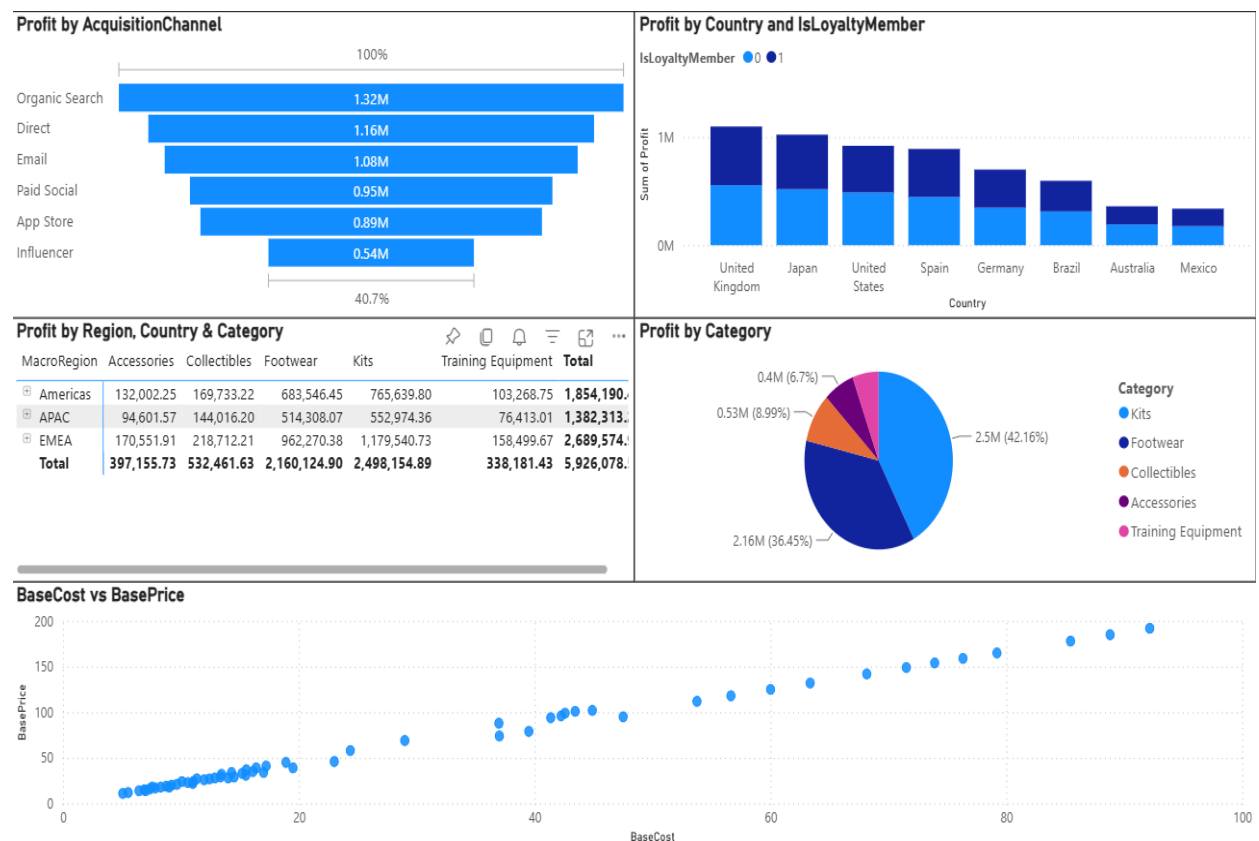


Fig. 2

Fig.2 from our Power BI dashboard shows the profits generated on our merchandize sales and gives us insight on what we can improve for better profitability.

Kits and accessories, footwear sold by Highbury are by far the highest generators of profit (~78%). EMEA (Europe, Middle East, Africa) accounts for the region generating the most profit which makes sense considering the club is based in London, but the Asia Pacific and Americas regions also contribute a significant portion to the profits (~53%) which shows that there is huge potential in these markets. Profit by Acquisition Channel shows that Organic Search was the highest, while profit through influencer collabs was the lowest as 0.54m pounds which calls for better social media marketing. The BaseCost vs BasePrice shows the cost relative to the selling price.

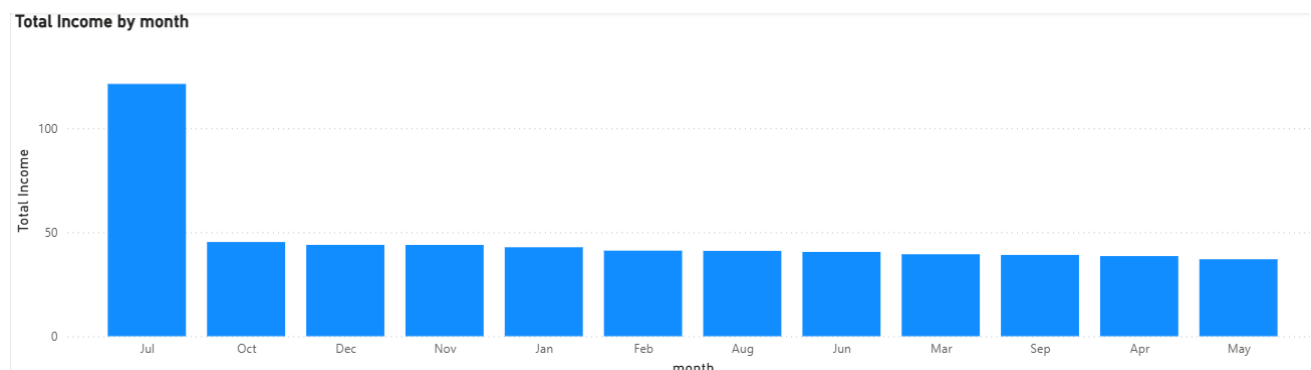


Fig. 3

Fig. 3 shows that July is the dominant income month (~121m pounds over the last 5 seasons), significantly higher than the other months because it lines up with pre-season tickets, memberships, kits sales and the transfer window is open.

(d) GAP/SWOT Analysis

SWOT

STRENGTHS	WEAKNESSES
• Historic club with a loyal fanbase. • Reputable youth academy, pure profit when players sold. • Financially stable and conservative with squad improving every year • Well established broadcasting channel	• No CRM, fan data left untracked • Not sufficient revenue from tickets/memberships/merch sales • Squad costs and wages more than 80% of total
OPPORTUNITIES	THREATS
• Growing sponsorship market in London • Digital fan engagement through socials, apps and email • Matchday hospitality packages • Youth academy expansion.	• Rival clubs competing for the same fanbase • High player wages and transfer market inflation • Lower fan spending power amid economic crisis • Income concentration during early season

GAP

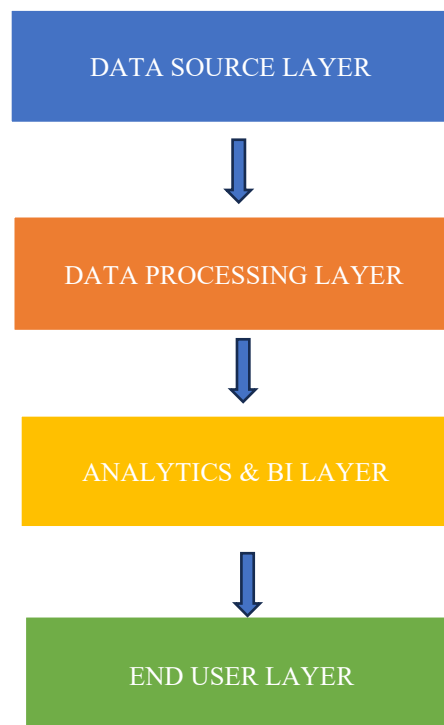
Analysis mapping the club's current state vs the requirements of at least a medium size club using BI and CRM solutions to address the gaps

Business Area	Current	Target	How gap is addressed
Fan Data	No CRM solution	Full CRM implementation	Salesforce CRM & fan engagement dashboard
Revenue	Only ~22% from commercial sales	40% revenue from commercial sales	Financial Dashboard and income tracking
Operational	Manual excel sheets	Real-time BI dashboards	Operational Dashboard – Spend & Inventory
Player records	Manual player performance records	Data driven insights	Player stats Dashboard
Digital Revenue	Digital assets unmonitored	Tracking digital sales and trends	AI dashboard for insights

Fan Data	Ticket resolution rate to be improved in CRM	From 33% to 75% resolution	Salesforce CRM
----------	--	----------------------------	----------------

(e) **System Design**

The BI & CRM solution for FC Highbury is a 4 tier architecture.



Data Source Layer

- Financial Transactions System – Club income and expense records from mockaroo
- Merch Sales System – Product, Order, Customer dataset (mock data from GitHub repo)
- Time Intelligence System – Date dimension table
- Inventory System – Inventory dataset from Kaggle
- Operations System – Operational transactions data from mockaroo
- Sorare NFT platform – Real data tracking monthly NFT sales from CryptoSlam extracted through API call
- Players Information System – Squad players statistics mock data
- Players Injuries Tracking System – Kaggle dataset containing PL players injury info.

Data Processing Layer

- Importing and validating CSV/XSLX files in Power BI Desktop

- Standardizing the data in terms of types, formatting, deleting unnecessary columns for a more efficient data model.
- Data modeling by setting up star schema between fact and dimension tables.
- Fin_data & fin_data_extra connected to DimDate via datekey for time intelligence
- Creating calculated columns & measures for KPIs like Revenue, Income, Expenses, Avg goals per 90, Injury risk score, etc.

Analytics & BI Layer

- Power BI Service – Publishing the dashboards from BI desktop.
- Salesforce CRM – Native list views, reports on objects contacts and opportunities.

End User Layer

- Management – Financial and Operational dashboards for strategic insights and decision making.
- Coaching Staff – Players dashboard for matchday tactics and selection.
- Marketing – Salesforce pipeline & AI dashboard for fan engagement analytics.
- Fans & Supporters – Indirectly through CRM workflows triggering actions and communication.

(f) Database Design

Entity Relationship Diagram

The financial dashboard data model can serve as an example of the ERD for this project as seen below.



Fig. 4

FactSales serves as the fact table while DimDate, DimCustomer, DimProduct, DimRegion serve as dimension tables in this star schema. These datasets are mock datasets from a GitHub repository /shannonlindsay/FabricCommunityContests. Fin_data & fin_data_extra are datasets from Mockaroo that contain the financial income and expenses data and the sorare_sales_monthly was extracted using an API from the CryptoSlam sports NFT site.

Data Dictionary – fin_data (Core table for financial data)

Field	Data Type	Description
Actual_Date	Date	Calendar date of the financial record
datekey	Integer	Numeric date key (YYYYMMDD) linking to DimDate
academy_income(m)	Decimal	Revenue from youth academy registrations and coaching fees (millions)
broadcasting_income(m)	Decimal	Revenue from TV and streaming broadcast rights (millions)
kits_accessories_income(m)	Decimal	Revenue from kit and accessory sales (millions)
loan_players_income(m)	Decimal	Income from player loan agreements (millions)
tickets_membership_income(m)	Decimal	Matchday ticket and membership subscription revenue (millions)
sponsor_deals(m)	Decimal	Income from commercial sponsorship agreements (millions)
player_sales(m)	Decimal	Transfer fee income from player sales (millions)
IT_services(m)	Decimal	Expenditure on technology and IT services (millions)
loan_repayment(m)	Decimal	Scheduled loan repayment expenses (millions)
misc_oper_expenses(m)	Decimal	Miscellaneous & operational expenditure (millions)
player_wages_bonuses(m)	Decimal	Total player wage bill including bonuses (millions)
player_purchase(m)	Decimal	Transfer expenditure on player acquisitions (millions)

share_dividends(m)	Decimal	Dividend payment to shareholders (millions)
staff_wages(m)	Decimal	Non-player staff wage expenditure (millions)

Datasets Summary

Table Name	No of Rows	Source	Description
fin_data	60	Mockaroo	Monthly financial income and expense data of FC Highbury over the seasons 21/22 to 25/26 and connected to DimDate by datekey for time intelligence.
fin_data_extra	60	Mockaroo	Similar to fin_data, added to include player_sales_income and player_purchase_fields that were left out in fin_data.
sorare_sales_monthly	81	Cryptoslam.io	Sorare cryptoslam data tracks NFT sales, transactions and unique buyer stats for the Sorare fantasy sports platform.
FactSales	133,362	Mock data from a open academic github repo	It is a fact table of merchandize sales and revenue
DimCustomer	30,000	Mock data from a open academic github repo	It is a dimension table of customer info that connects to the FactSales via a one-to-many relationship using CustomerKey
DimProduct	100	Mock data from a open academic github repo	It is a dimension table of merch info that connects to the FactSales via a one-to-many relationship using ProductKey
DimRegion	8	Mock data from a open academic github repo	It is a dimension table of customer region info that connects to the FactSales via a one-to-many relationship using RegionKey
DimDate	1826	Mock data from a open academic github repo	It is a date table spanning from 2021 to 2025 for time intelligence calculations and visuals
players	28	Mockaroo	Mockaroo dataset containing player game stats like goals, assists, xG, position, release_clause, training_rating, minutes_played, interceptions, tackles
Player_injuries_impact	656	Kaggle	Kaggle dataset containing premier league players injury records, type, date of injury, date of return, matches missed, etc.,
Fact_OperationalExpenses	720	Mockaroo	Mockaroo dataset generated to account for operational expenses at the club such as

			electricity, fuel, legal fees, insurance, consulting fees, visa immigration, EMS, scouting fees, etc.,
Fact_Supply_Orders	158,027	Kaggle	Kaggle dataset containing stadium supplies, club equipment, training kits supplier info, wuantities, delivery days, quantity ordered, etc.,

Implementation Report

Overview

This implementation report reflects on the creation of the business intelligence and CRM solution for the football club, FC Highbury. The organization is a lower-league soccer team that undergoes a management transition and plans to scale its operations in the future. The initial assessment revealed that the club lacked ticket sales, had no digital presence, and did not use any customer relationship management (CRM) software to track its operations and customers' behaviors. This report illustrates how these issues were resolved by implementing Salesforce's customer relationship management system and three Power BI dashboards, including Team, Fans & Operations, and Management (Financials). The creation process of the selected tools will be discussed, and a comparison with the fourth dashboard, created with artificial intelligence (AI) and based on the same data, will be provided.

(a) Development Process

The club's initial model did not account for the squad's injuries and risk assessment. The proposed initial design tackled this issue by combining two sets of data: The club's squad and players' statistics from last season and a similar set of data from the Premier League used as a reference to evaluate the risk of injuries [1]. Since there was no overlap between key indicators for the two sets, they were joined using playing positions as a bridge.

The external injury dataset required extensive cleaning due too several issues. There were inconsistencies in position labels, some ratings were recorded as text to flag substitute-appearance (e. g. "6(S)"), there were also some missing values and typos on the records of the dataset and open injuries recorded with a placeholder "Present" rather than a return date. All were resolved by changing them to proper null instead of a fabricated number, leaving the subsequent averages unaffected.

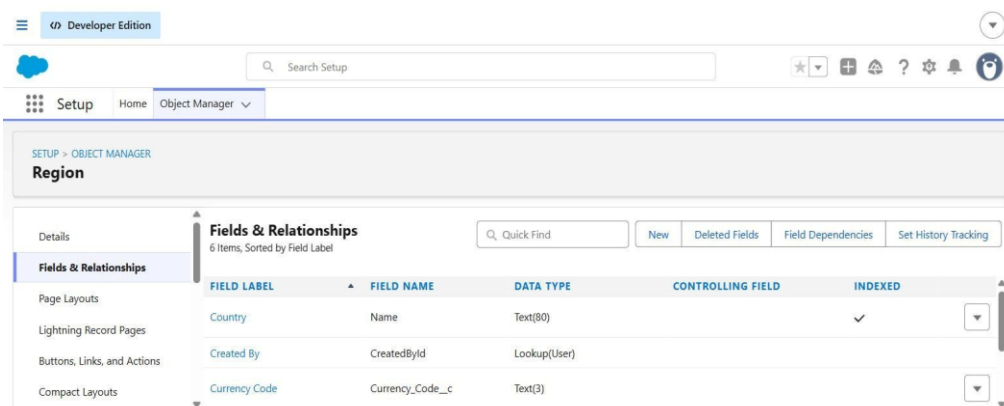
The most significant correction had to be made when constructing the Injury Risk Score itself. To begin with, the positional injury-frequency statistic was weighted in the calculation as if it had a scale from 0 to 1, but in fact, upon examination, it turned out that the frequency for each position varied from 1.5 to 4.0. This means that the weighting factor alone would have already given all players a "high risk" category. This method was corrected by normalizing the value using the minimum and maximum values found so that it fits the 1 to 0 scale, after which it was used in the weighted mean.

Similarly, on the CRM side, an iterative approach was taken as well. During the testing process, reference data (e. g. the Region list) was discovered to have one record duplicated, which was then removed, and the initial merge-field configuration on the High-Value Booking Alert was corrected, because testing had revealed that an invalid token was present in the message. Again, just like with the dashboards, these were unintended discoveries made during testing that were then resolved.

As a part of the search for additional revenue streams, one option was to create an NFT-based fan engagement channel (i.e., a digital collectible card). Sales data was analysed in detail to understand the trend, which revealed that the value first sharply increased in 2021-2022 but then decreased significantly, which is evident in the Fig. Based on these findings, the team concluded that the NFT market was not a sustainable source of income. Thus, this option was not included among the recommended solutions, and the analysis was a part of the final justification report.

(a) CRM System Implementation

The CRM was deployed in Salesforce to solve the lack of customer relationship management and the fan data tracking system in the club. There are four major objects in the system: the Region, Member, Matchday Hospitality Booking, and Ticket. The first object refers to the fan's location or area with six fields, including the currency code (for multiple currencies reporting). The second object deals with the data on the fans or customers, with twelve fields, two of which are the Acquisition channel and the Date of Birth. The third object stores the booking data on members for hospitality on match days with eleven fields, including the booking reference number (auto-number) and the booking status (picklist). The fourth object records the ticket-related service requests to track the SLA. A member can make multiple bookings and tickets but only belongs to one region. This is a one-to-many lookup relationship that avoids double-counting entries and facilitates roll-up reports from the related objects (see figures 1, 2, and 3 below).



FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Country	Name	Text(80)		✓
Created By	CreatedById	Lookup(User)		
Currency Code	Currency_Code_c	Text(3)		

Figure 1: Custom fields on the Region object, including the Currency Code field.

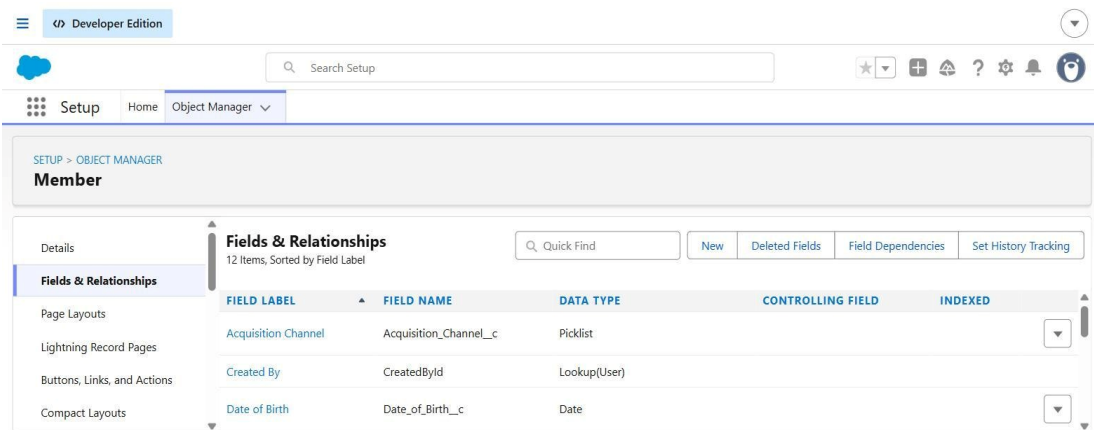


Figure 2: Custom fields on the Member object, including Acquisition Channel and Date of Birth.

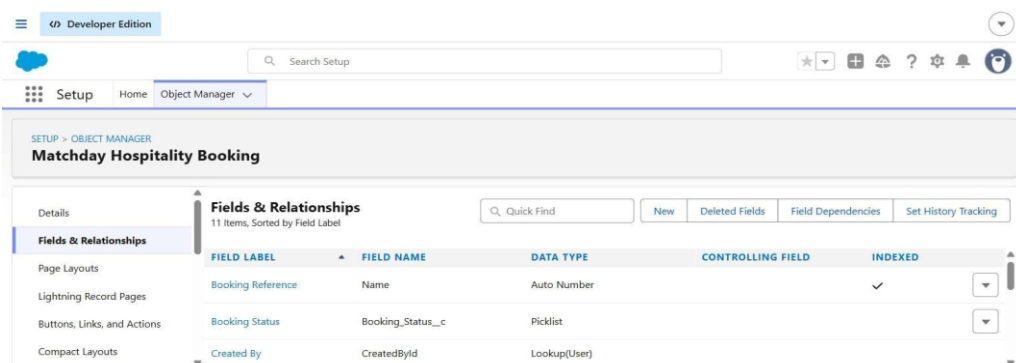


Figure 3: Custom fields and relationships on the Matchday Hospitality Booking object.

Data integrity is enforced by validation rules on the objects that most require data entry controls. Firstly, the Booking object has a rule that sets the number of guests that can be set from one to twenty, called `Guest_Count_Reasonable`, and a rule that does not allow to confirm booking for a match that has already happened, called `No_Confirming_Past_Matches` (Figure 4). Secondly, the Member object has `DOB_not_future` and `Join_Date_Not_Future` rules in order to prevent entering of future dates when a member's date of birth or joining date is needed (Figure 5).

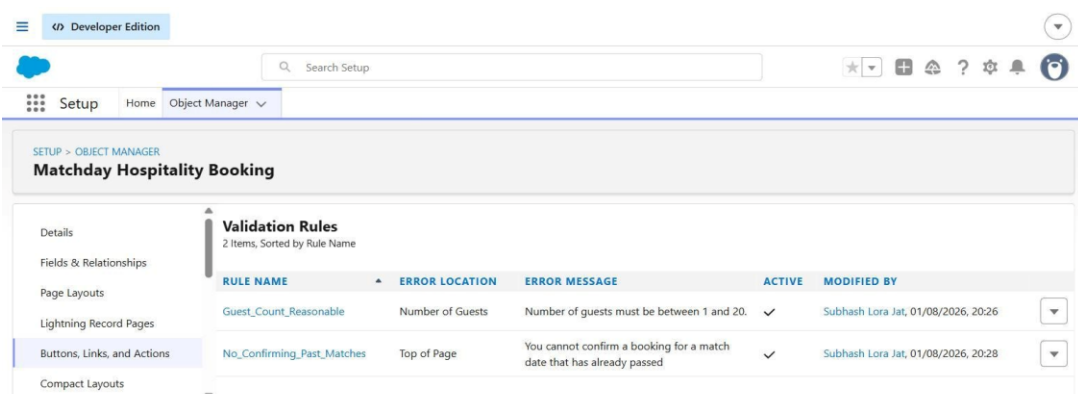


Figure 4: Active validation rules on the Matchday Hospitality Booking object.

Developer Edition

Search Setup

Setup Home Object Manager

SETUP > OBJECT MANAGER

Member

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Validation Rules
2 Items, Sorted by Rule Name

RULE NAME	ERROR LOCATION	ERROR MESSAGE	ACTIVE	MODIFIED BY
DOB_not_future	Date of Birth	Date of Birth cannot be in the future.	✓	Subhash Lora Jat, 01/08/2026, 20:31
Join_Date_Not_Future	Join Date	Join Date cannot be a future date.	✓	Subhash Lora Jat, 01/08/2026, 20:32

Figure 5: Active validation rules on the Member object.

Declarative automation, which takes the form of Salesforce Flows in Salesforce, can replace long and complex processes that would otherwise require manual staff intervention. Firstly, a Booking Confirmed autolaunched flow is triggered by the status change. Secondly, a more complex High Value Booking Alert flow is triggered on record creation or update and assesses the booking's value; if it exceeds a certain threshold, a follow-up task is created for the user, as well as a chatter feed post (Figure 6), which serves as an alert mechanism, notifying the staff in a timely manner regarding the high-value booking, eliminating the need for manual checking of the booking list.

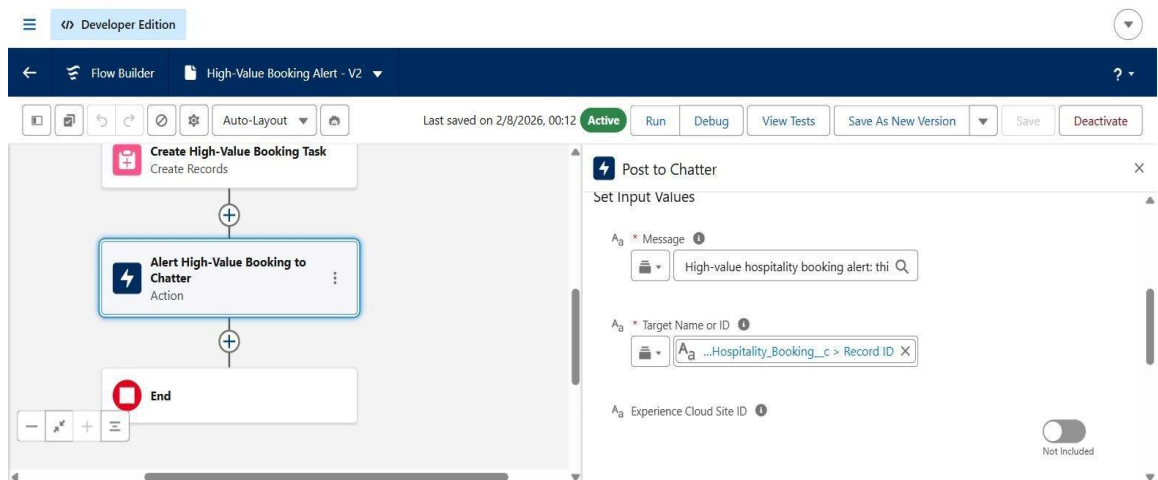


Figure 6: High-Value Booking Alert flow — automated Chatter notification action.

This automation is related to chatter group called Match Day Hospitality Team which is used for internal communication. The results of this automation can be seen on the record in Figure 7 below which shows a confirmed booking with the automatic update being seen in its chatter feed, therefore showing the process from start to finish.

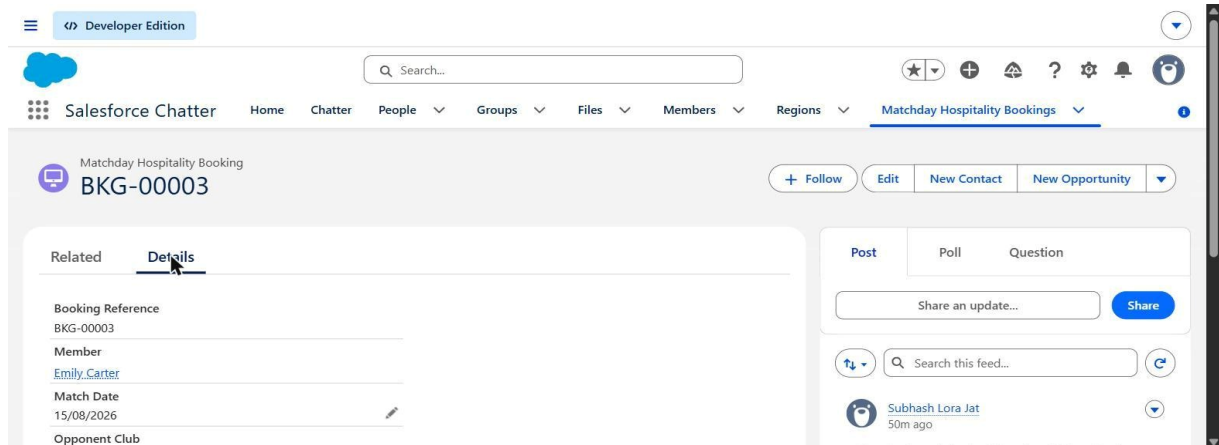


Figure 7: Booking record layout showing the automated high-value alert posted to Chatter.

Filtered list views for Bookings, Members, and Regions were set up to ensure day-to-day operations, and three summary reports on Bookings by Package Tier, Members by Region, and SLA Compliance by Month were combined into one operational dashboard (Figure 8). The dashboard provides an insight into sales mix, fan distribution, and SLA compliance for the hospitality and membership departments. Notably, the improvements in the SLAs were consistent across all months, with the number of resolved tickets within the SLA scope increasing from 3 out of 9 in January to 9 out of 9 in April, and the average resolution time down by 27.1 to 8.6 hours, which was also reported by the AI-generated dashboard used in the comparative analysis below.

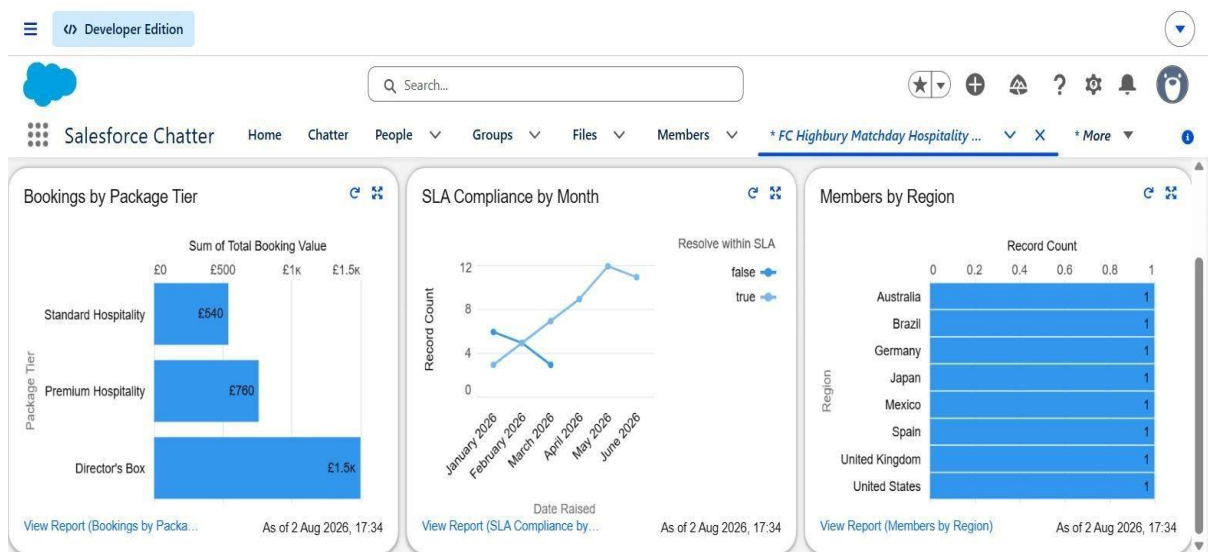


Figure 8: FC Highbury Matchday Hospitality dashboard, combining booking, membership, and SLA reports

(b) Dashboards & Reports

Team Dashboard

The Team dashboard aims to address the problem of the club’s lack of information about the squad’s performance and potential injuries. It uses the club’s data about the squad and the external injury dataset matched via a bridge table that aligns players’ positions. This allows for calculating each player’s Injury Risk Score, which is based on empirical data rather than estimations. Figure 9 below shows the Injury Risk page, which contains the list of players ordered by their Injury Risk Score and Tier, a scatter plot that visualizes the risk vs. market value of players to identify high-value and high-risk players, and a bar chart that displays the underlying position’s benchmark resulting in the scores.

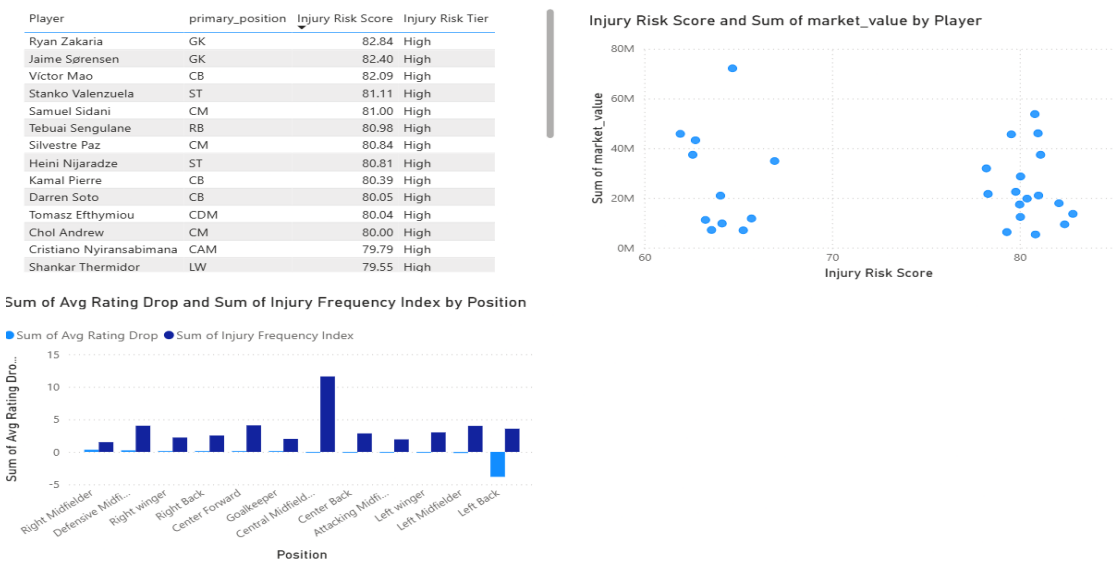


Figure 9: Team dashboard — Injury Risk page.

A Performance page (Figure 11) allows distinguishing between the core of starting XI and the rotation by analysing the amount of minutes per game, and an in-depth player on-ball statistics table (Figure 12) displaying touches, passes, tackles, duel success rate, and other stats sorted by positions is available for further exploring.

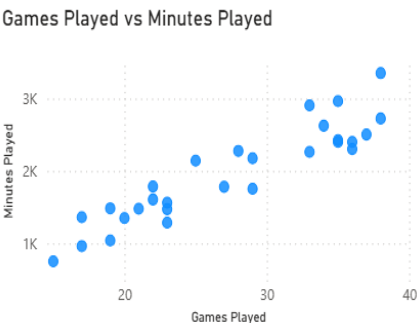


Figure 11: Team dashboard — Performance page, minutes played vs games played.

Player On-Ball Statistics

Player	Touches	Passes	Tackles	Duel Success
Arkadiusz Umar	2017	1196	6	59.10
Arseniy Laabidi	2206	1671	24	71.50
Bo Narine	1010	752	0	54.60
Chol Andrew	2466	2495	20	60.80
Claudio Makoond	2412	1587	35	82.50
Cristiano Nyiransabimana	2837	1674	13	48.90
Darren Soto	1566	1556	22	72.60
Daryl Nkrumah	2364	1329	51	66.10
Dimitriy Gerasimov	2112	1433	12	57.00

Figure 12: Team dashboard — player on-ball statistics table.

Fans & Operations Dashboard

This dashboard covers the specification's 1st Operational gap and contributes to the 2nd Revenue gap by addressing commercial profit. It responds to the club's need to assess the lack of operational cost control and fan/customer engagement tracking. Figure 13 below shows the distribution of all operating spend, with the breakdown per category revealing that electricity, scouting fees, and stadium rent, and lease constitute the largest expenditure items. This information was useful for the club's stated intent to diversify suppliers to obtain better commercial terms. Another page (Figure 14) was contributed to the report showing merchandise and fan-engagement profit by acquisition channel, country, customer loyalty, and category, which supports the club's need to develop merchandising and engage more customers.

Club Operating Spend Dashboard - Inventory Procurement vs Operational Expenses

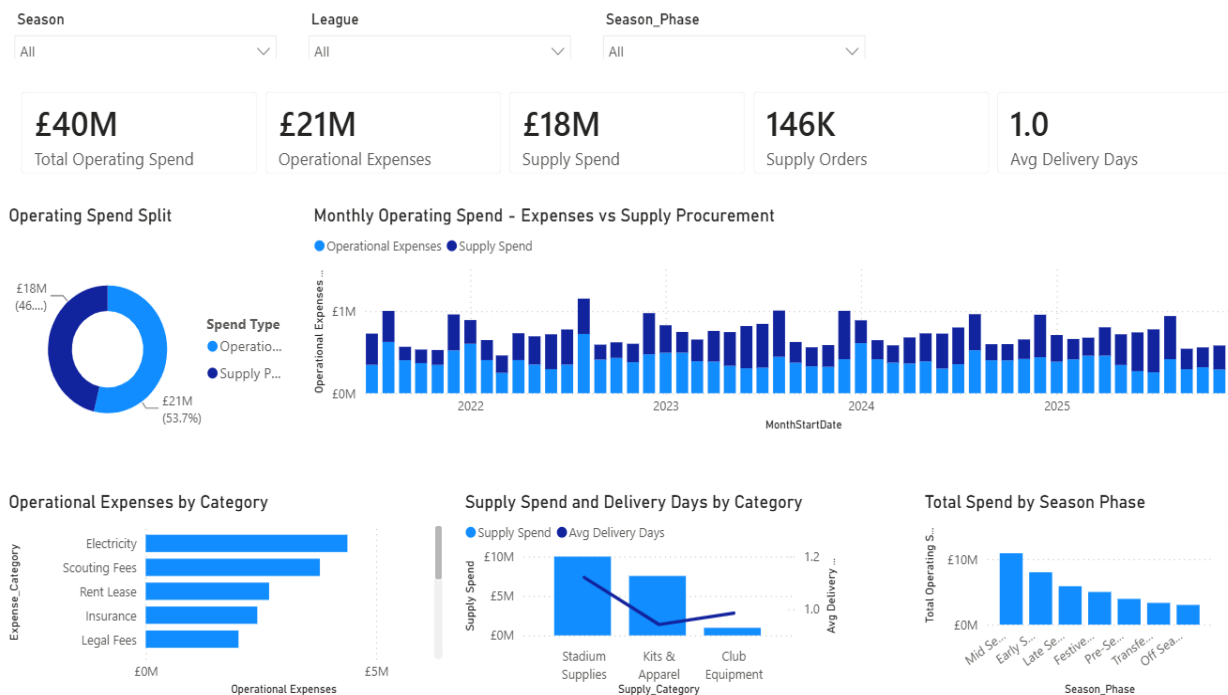


Figure 13: Fans & Operations dashboard — operating spend by category and supply chain.

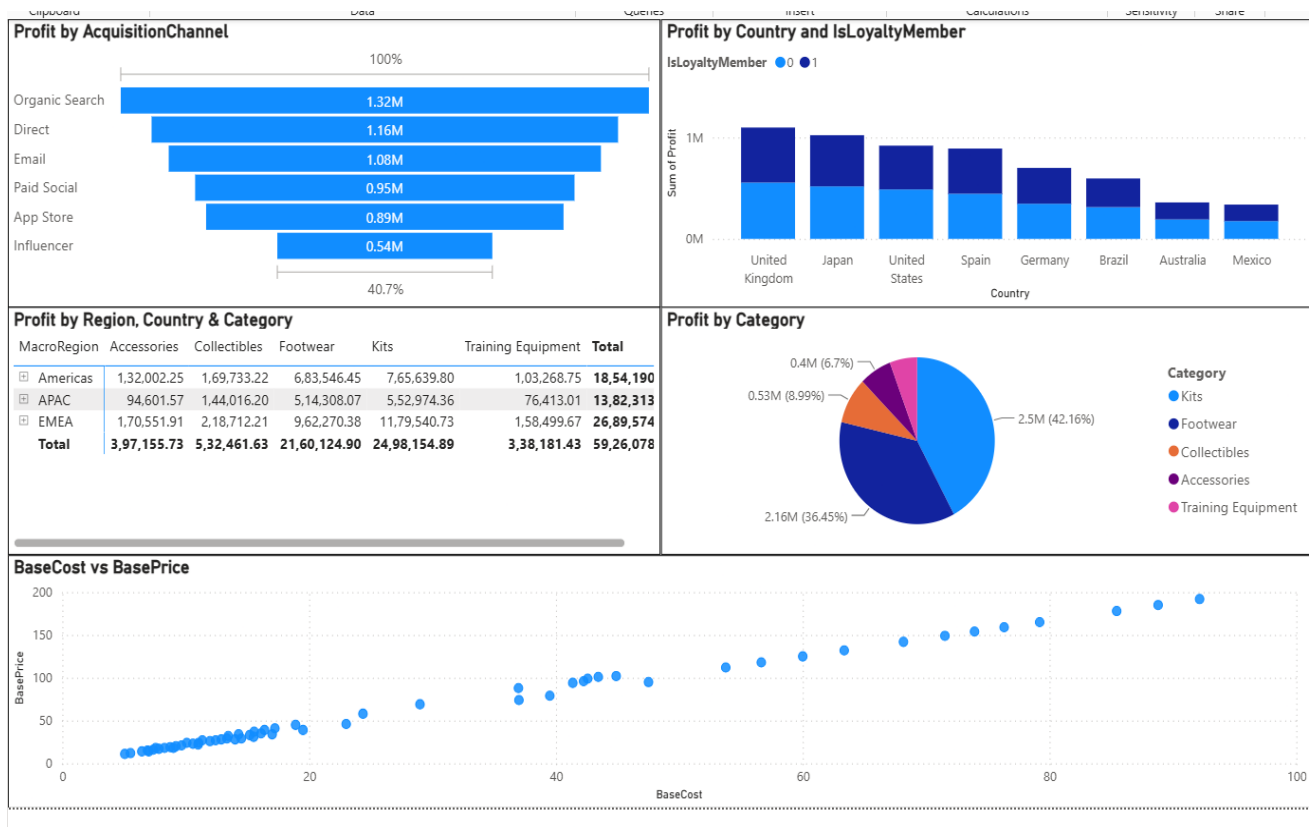


Figure 14: Fans & Operations dashboard — profit by acquisition channel, country, and category.

Comparative Analysis: AI-Generated vs Originally Created Dashboards

This section also covers the last identified weakness in the analysis of the specification – Digital Revenue, where prior to the improvements, the digital assets were not tracked. Firstly, a fourth dashboard was designed using the AI tool which aggregated the retail sales, club finances, squad, membership, and CRM data, providing a single pane of glass view of the information (Figure 13) to compare it to the three dashboards created manually above.



Figure 13: AI-generated dashboard — header, headline KPIs, and revenue forecast.

The dashboard finally produced forecast figures that were not found in the manual Excel workbook at all (Figure 14). It includes a revenue forecast based on the compound annual growth rate (CAGR) of 9.4%, which is projected to reach £2.84M in 2026 and £3.11M in 2027. Additionally, there is a linear projection of the squad's value, which, assuming the current investment level, is expected to reach £415.5M in 2027/28.



Figure 14: AI-generated dashboard — revenue forecast and product category mix.

Some of the AI dashboard's figures can be verified by the manually built dashboard. In particular, the CRM panel (Figure 15) shows that SLA compliance increased from 33% to 100% during the six-month period while the average resolution time decreased from 27.1 to 8.6 hours, which is exactly the same as the figures from the manually built dashboard's SLA Compliance report (despite it being built on the ticket data). Therefore, one can conclude that this AI dashboard has correctly aggregated the source data rather than simply guessing the figures.

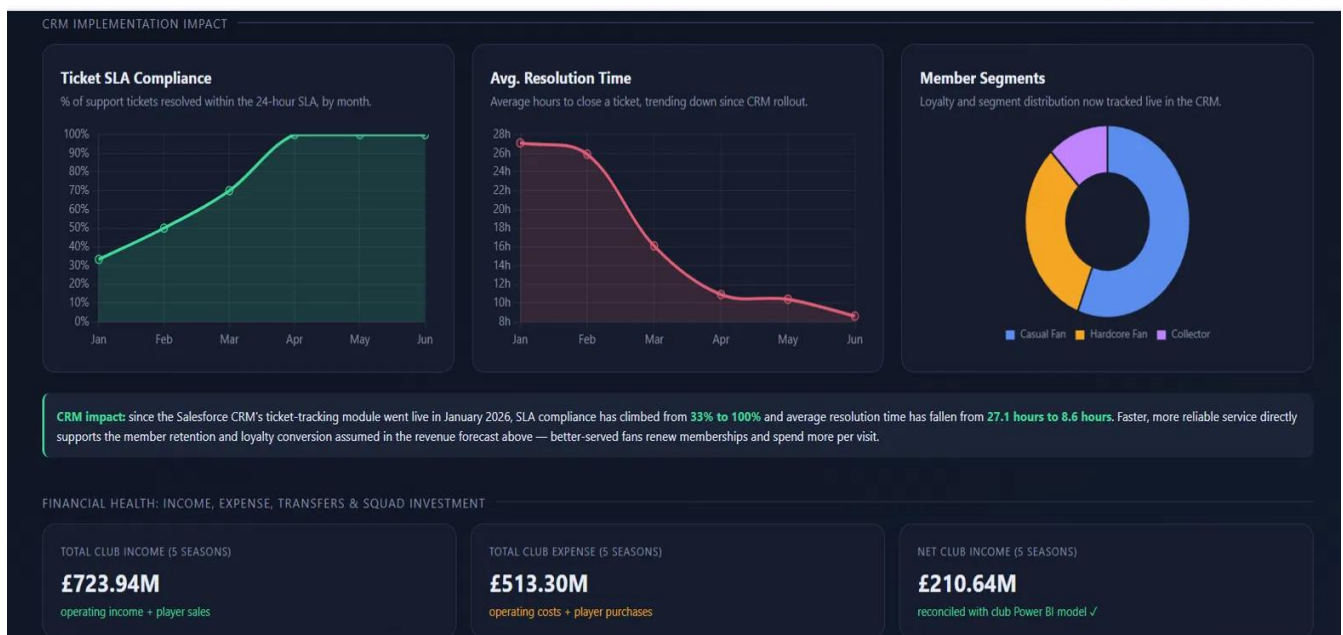


Figure 15: AI-generated dashboard — CRM implementation impact panel.

This is also an example of something the AI dashboard was able to identify that the manually-created dashboards were not: specifically, an explanation of why there were two different reported figures for ‘net income’ in the data set. The auto-generated summary in Figure 16 highlights that there are actually two valid ways to calculate net income in this particular data set – one representing a rolling, year-round figure for the whole club (£15.97M for the 2025 TYE, down 70.9% y-o-y due to extensive transfer spending), and the other a per-season figure based only on revenues (£167.78M for the 2025/26 season, up 49.3% compared to 2021/22). The summary reason for this apparent discrepancy, and an explanation as to why both figures are in fact accurate within their respective contexts, is provided. Additionally, it was only through the AI dashboard that the insight that net club income had been falling every season since 2023/24 and turned negative (-£5.4M) in the current season as a consequence of the aforementioned transfer policy could be identified – the manually-created dashboards, built as they were as three separate excel files by different team members, do not cross-reference each other or perform this specific calculation.

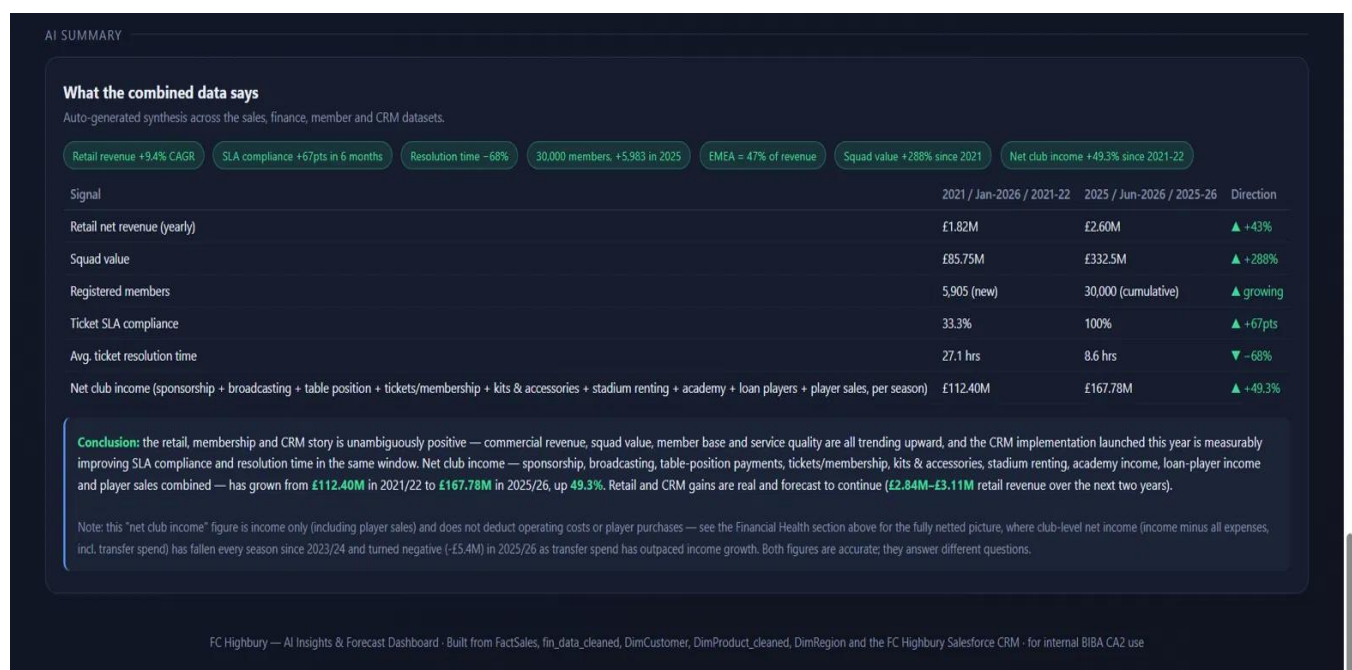


Figure 16: AI-generated dashboard — auto-generated cross-dataset summary and conclusion.

In general, it can be concluded that the AI-created dashboard was easier to create and was better in terms of combining different sets of data, as can be seen from the fact that net-income reconciliation appeared as a helpful insight. However, the manually created dashboards contain several data quality improvements noted in section (a) that could not have been made if the AI dashboard was used due to the lack of transparency in its workings. Therefore, both methods should be used in the future as the AI method lacks judgment in specific areas, including reconciling the data sets, while manual creation might take too much time searching for patterns for the AI to do itself.

(c) Initial Response to the Deployed Solution

Data tracking and CRM

Prior to this project, the club did not have an organized database of its clientele, a systematic approach to booking hospitality, or a way to track response times. After implementing the CRM, the club now has a unified view of Member and Booking records, an automated high-value booking alert system that allows the hospitality team to respond faster to high-value sales opportunities. The improvement in SLA compliance, which increased from approximately one-third to full compliance (per the independent verification by the AI dashboard), is indicative of the positive effect on the club's ability to provide fast customer support.

Financial and operational visibility gap

Prior to this project, there was no aggregated view of cost drivers and revenue sources. As the Fans & Operations dashboard below demonstrates, after implementing the solution, the largest costs are electricity, scouting fees, and stadium rent or lease. The specification report's Management dashboard is an aggregated view of the club's income and expenses, while the comparative analysis above provides another valuable insight not available before. Specifically, the Financial Health panel in the AI-generated dashboard indicates that the club's net income has decreased for two consecutive seasons and become negative in 2025/26 due to increased spending on transfers. This is a critical discovery that the club's board had not been aware of and that directly informs the need for the supplier cost-reduction and revenue diversification initiatives identified in the specification.

Squad management gap

Before this project, the risk of injuries was evaluated in an informal manner. Now, after the implementation of the Team dashboard, the staff can recognize the players that are both high-value assets and have injury risks, which previously had gone unnoticed, exposing the team to potential losses.

Limitations of this initial response

This assessment is still a projection because the recommendation has yet to be tested on a live club season. Injury Risk Score is still a modelled proxy based on positional averages in spite of being normalised, not individualised according to medical history. The AI dashboard's suggested CAGR is a linear extrapolation of 5 years of data and should be regarded as a directional guide, especially on a negative net income trend line.

(d) Reflection & Future Work

Reflection

The most challenging aspect of this implementation was debugging the reason behind the consistent “High” outputs for the Injury Risk Score measure. Despite a superficially logical underlying DAX formula, it failed to account for the metric’s domain logic due to a hidden inconsistency in calculations. Specifically, the positional injury-frequency multiplier always scored above the “high risk” threshold due to the value range of this multiplier implicitly assuming a 0-1 interval for injury frequency, whereas in reality, it varied between 1.5 and 4. This domain inconsistency was debugged by reviewing the distribution of the injury-frequency measure and hypothesizing about potential value ranges – ideally, one would not have to “guess” the potential value ranges for a metric, but rather design its calculation logic around a known domain.

The second important lesson was the need for a reconciliation process between the data models. When building dashboards in parallel, it is entirely possible to implicitly accept minor inconsistencies that would be immediately visible if one had to operate within a single dashboard (for example, there being two different formulas for calculating net income across dashboards). While in this particular case, an automated AI reconciliation resolved the issue, it is evident that prior to the AI intervention, no such reconciliation process existed between dashboards. This suggests that, even if no AI tools were used, the team should have implemented a manual reconciliation procedure before finalizing the dashboards to ensure consistency.

Future Work

The future work on the project should be focused on several areas. First, the priority must be set on incorporating individual medical history, once available, in order to obtain a more relevant injury risk score. Second, the use of CRM automation to manage renewals of season passes and memberships should be considered in order to promote the club’s goals of retaining existing customers. Third, it is recommended to establish a reconciliation procedure between dashboards as a closing procedure to ensure that definitions are consistent across different dashboards, as was revealed in the current project by the anomaly in the AI dashboard. Finally, the idea of using NFT sales as a growth channel should be re-evaluated if market analyses reveal a downward trend in their demand in the future.

References

- [1] [A. Biswas, “Player Injuries and Team Performance Dataset,” Kaggle.](#)
- [2] <https://www.kaggle.com/datasets/gabrielsantello/wholesale-and-retail-orders-dataset>
- [3] <https://github.com/shannonlindsay/FabricCommunityContests/tree/main/StarterFiles>
- [4] <https://www.cryptoslam.io/sorare/sales/summary>
- [5] <https://www.deloitte.com/uk/en/services/consulting-financial/analysis/deloitte-football-money-league.html>

[6] <https://www.pwc.ch/en/insights/sport/sports-survey-2023.html>

Appendix

Appendix A: Data Sources

Source	Type	Used in
Squad database (28 players)	Mockaroo Generated	Team Dashboard
External injury dataset (real top-flight match records)	Benchmark Data from Kaggle	Team Dashboard – Injury Risk Page
Retail sales, customer, product, region, and calendar dimension tables	Mockaroo Generated	Fans & Operations dashboard; AI-generated dashboard
FC Highbury Salesforce CRM (Bookings, Members, Regions, Tickets)	System of record	CRM system implementation; AI-generated dashboard

Appendix B: Key DAX Measures (Team Dashboard)

Injury Risk Score — combines a normalised positional injury-frequency benchmark with player-level fitness and current-injury indicators:

```
1 Injury Risk Score =
2 VAR PositionFreq =
3     CALCULATE(
4         MAX(InjuryBenchmark[Injury Frequency Index]),
5         USERELATIONSHIP(Players[Primary Position], PositionBridge[primary position])
6     )
7 VAR MinFreq = CALCULATE(MIN(InjuryBenchmark[Injury Frequency Index]), ALL(InjuryBenchmark))
8 VAR MaxFreq = CALCULATE(MAX(InjuryBenchmark[Injury Frequency Index]), ALL(InjuryBenchmark))
9 VAR PositionFreqNormalized = DIVIDE(PositionFreq - MinFreq, MaxFreq - MinFreq)
10 VAR FitnessPenalty =
11     DIVIDE(100 - SUM(Players[vo2_level]), 100) * 30
12 VAR BmiPenalty =
13     DIVIDE(ABS(SUM(Players[bmi]) - 22), 22) * 20
14 VAR CurrentInjuryPenalty =
15     IF(SUM(Players[Weeks Out]) > 0, 15, 0)
16 RETURN
17     (PositionFreqNormalized * 50) + FitnessPenalty + BmiPenalty + CurrentInjuryPenalty
```

Injury Risk Tier — converts the score into a High/Medium/Low band for use in conditional formatting and the risk-ranked table on the Injury Risk page.

```

1 Injury Risk Tier =
2 SWITCH(TRUE(),
3     [Injury Risk Score] >= 70, "High",
4     [Injury Risk Score] >= 40, "Medium",
5     "Low"
6 )
7

```

Appendix C: CRM Object Summary

Object	Fields	Key relationships
Region	6 fields, incl. Currency Code	Parent to Member
Member	12 fields, incl. Acquisition Channel, Date of Birth	Belongs to Region; parent to Booking and Ticket
Matchday Hospitality Booking	11 fields, incl. Booking Reference, Booking Status	Belongs to Member
Ticket	Used for SLA compliance reporting	Belongs to Member

Work Split

FC Highbury — Business Intelligence & Business Analytics Project

Task Allocation

Team Member	Contribution
Arun	Implementation Report; Team (Player) Dashboard
Subash	CRM Implementation; AI-Generated Dashboard; Fans & Operations Dashboard
Cariappa	Financial (Management) Dashboard; Specification Report

Workload Split

Team Member	Share of Total Work
Arun	33.3%
Subash	33.3%
Cariappa	33.3%

Tasks were divided approximately equally across the three team members.